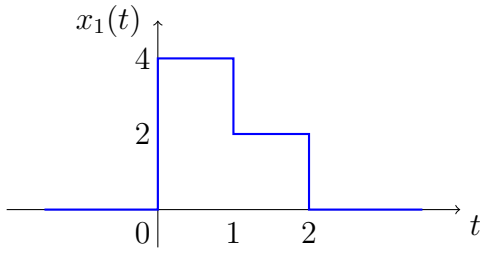
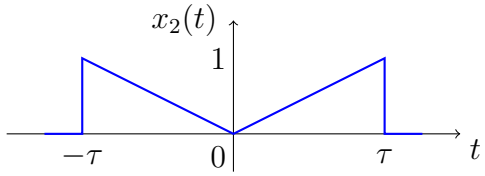
EE1101 Signals and Systems JAN—MAY 2020
Tutorial 7

1. Find the Fourier Transform of the following signals. Use $\text{sinc}(\omega) = \frac{\sin(\pi\omega)}{\pi\omega}$ to simplify the answer.



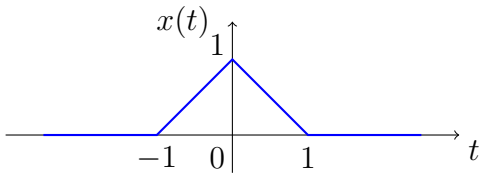


2. Use the Fourier Transform analysis equation to calculate Fourier Transform of the following signals
   (a) $\delta(t+1) + \delta(t-1)$
   (b) $\dfrac{d}{dt}\{u(t-2) + u(-2-t)\}$
   Sketch and label the magnitude spectrum for both parts (a) and (b).

3. For each of the following signals $x(t)$, compute the Fourier transform $X(j\omega)$:
   (a) $x(t) = e^{-\frac{|t|}{2}}$
   (b) $x(t) = (\sin 2\pi t) \cdot e^{-t}u(t)$

4. Consider the signal
$$x(t) = \begin{cases} 0, & |t| > 1 \\ \dfrac{t+1}{2}, & -1 \le t \le 1 \end{cases}$$
   (a) Determine the Fourier Transform of $x(t)$.
   (b) Sketch the even part of $x(t)$ and find its Fourier Transform. Check if it is equal to the real part of your answer to part (a).
   (c) What is the Fourier Transform of the odd part of $x(t)$?

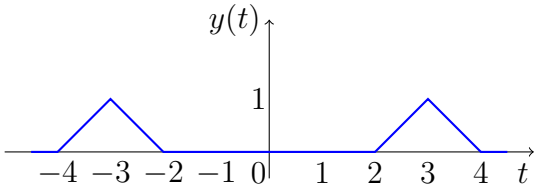
5. (a) Find the Fourier Transform of the signal $x(t)$ shown below.



(b) For any arbitrary signal $x(t)$, show that if $x(t) \Longleftrightarrow X(j\omega)$, then

$$x(t+T) + x(t-T) \Longleftrightarrow 2X(j\omega)\cos\omega T$$

(c) Using the above result, find the Fourier Transform of the signal $y(t)$ shown below.



6. (a) Show that if $x(t)$ is an even function of t, then

$$X(j\omega) = 2\int_0^\infty x(t)\cos\omega t dt$$

(b) Show that if $x(t)$ is an odd function of t, then

$$X(j\omega) = -2j\int_0^\infty x(t)\sin\omega t dt$$

7. Use the Fourier Transform synthesis equation to determine the inverse Fourier Transforms of:

(a) $X_1(j\omega) = 2\pi\delta(\omega) + \pi\delta(\omega - 4\pi) + \pi\delta(\omega + 4\pi)$

(b)

$$X_2(j\omega) = \begin{cases} 2 & 0 \le \omega \le 2 \\ -2 & -2 \le \omega < 0 \\ 0 & |\omega| > 2 \end{cases}$$

— END —